

TCP Congestion Solved Problems

Problem#1

Consider a TCP connection using slow start and congestion avoidance.

- The initial slow-start threshold (ssthresh) is 15 MSS.
- The congestion window (cwnd) changes over the rounds as shown:

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13
cwnd (MSS)	1	2	4	8	15	16	17	18→9	10	11	?	13→1	2	4

- I. Identify why **cwnd = 15** at round 4.
- II. What will be the **cwnd** at round 10?
- III. Between **Round 7 and 11**, which one indicates more severe congestion?

Solution:

- I. cwnd touches the threshold
- II. 12
- III. 11 as timeout occurs on this round

Problem#2

A file transfer starts at cwnd=1 MSS; ssthresh=9; round=0. The system faces a time-out at around 6 and then a 3-dup-ack on round 11.

- Identify the cwnd size on round 15.
- Identify the ssthresh on round 15.
- Total of how many bytes were successfully sent till round 15 if each MSS held 3 bytes of data?

Solution:

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
cwnd	1	2	4	8	9	10	11->1	2	4	5	6	7->3	4	5	6	7	

Ssthresh=5

Ssthresh=3

1. Cwnd on round 15: 7
2. Ssthresh on round 15: 3
3. Total bytes up to round 15: $3 \times (1+2+4+8+9+10+1+2+4+5+6+3+4+5+6+7)$

Problem#3

1. In TCP Congestion Control, the sender started (at round=0) with the slow start and the initial ssthresh was set to 16. Then the following events occurred.

- Event 1: Time-out occurred at round = 9
 - Event 2: 3 Duplicate acks received at round= 16
 - Event 3: 3 Duplicate acks received at round= 20
 - Event 4: Time-out occurred at round = 28
- a. Draw the cwnd vs round graph for the above scenario.
 - b. Calculate the value of ssthresh at round=35.
 - c. Calculate the value of cwnd at round=35.

Solution:

Ssthresh=16					Ssthresh=10								Ssthresh=6							
Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
cwnd	1	2	4	8	16	17	18	19	20	21->1	2	4	8	10	11	12	13->6	7	8	9

Ssthresh=5					Ssthresh=6											
Round	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
cwnd	10->5	6	7	8	9	10	11	12	13->1	2	4	6	7	8	9	10

- a) Draw the figure following the above data. Assume round as the x-axis while cwnd as the y-axis.
- b) 6
- c) 10

Problem#4

A TCP connection initiates at Round 0 with a cwnd of 1 MSS and a ssthresh of 18 MSS. By Round 15, the cwnd is observed at 11 MSS with an ssthresh of 7 MSS. It is noted that a 3-duplicate ACK event occurred at Round 7.

Based on these parameters:

- (a) Determine the value of ssthresh immediately following the event at Round 7.
- (b) Determine if a subsequent congestion event occurred between Round 7 and Round 15. Justify the findings briefly.
- (c) Identify the specific round in which this secondary event occurred.
- (d) Identify the type of congestion event (Timeout vs. Triple Duplicate ACK) that took place.

Solution:

Let's try to trace the cwnd over round

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Cwnd	1	2	4	8	16	18	19	20- >10								11
Ssthresh=18								sssthresh=10				sssthresh=7				

- a) Since 3-duplicate ACK occurs on round 7, $\text{ssthresh} = 20/2 = 10$.
- b) Ssthresh on rounds 7 and 15 are 10 and 7, respectively. Since a ssthresh changes if any congestion event occurs and ssthresh on rounds 7 and 15 are different, there must have at least one congestion event between the rounds.
- c) Let's backtrack from the round 15. On round 15, $\text{cwnd} > \text{ssthresh}$, so it is CA. thus decrease cwnd by 1 on each round.

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Cwnd	1	2	4	8	16	18	19	20- >10				7	8	9	10	11
Ssthresh=18								sssthresh=10				sssthresh=7				

On round 11, $\text{ssthresh} = \text{cwnd}$. This can happen in two cases: i) transition from slow-start to CA and ii) 3-duplicate ACK. Let's investigate what happened on round 10.

i) A transition from SS to CA occurs on round 10, the progression would look like as follows:

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Cwnd	1	2	4	8	16	18	19	20->10	1	2	4	7	8	9	10	11
Ssthresh=18								ssthresh=10			ssthresh=7			ssthresh=7		

In this case, the cwnd aligns perfectly. However, between round 8 and 15, we see two different ssthresh. However, either a time-out or a 3-duplicate ACKs event occurs. Hence, the assumption of the slow-start phase before round 11 is not valid.

i) Assume that there was a 3-duplicate ACKs on round 11. Then, cwnd progression look like this:

ii)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Round																
Cwnd	1	2	4	8	16	18	19	20->10	11	12	13	14->7	8	9	10	11
Ssthresh=18								ssthresh=10			ssthresh=7					

Now, the cwnd is aligned and a change of the ssthresh on round 11 is observed.

Thus, on round 11 there is 3-duplicate ACKs event.

d) 3-duplicate ACK